

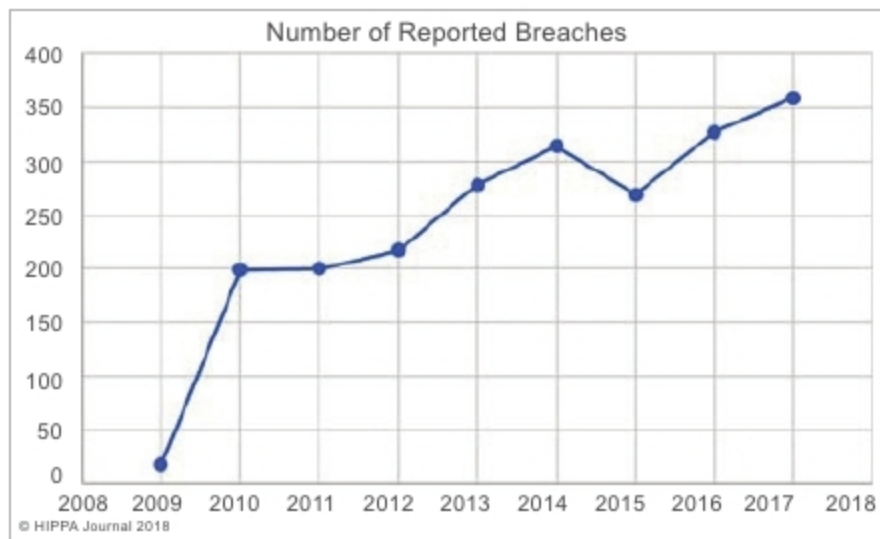


Figure 3: Statistics of the increase in the security breaches in the healthcare sector (Image source: [googleimages.com](https://www.googleimages.com))

before it gets a chance to interact with the application.

Web app firewalls use several different heuristics in order to determine which traffic is given access to the app and which needs to be removed. A constantly-updated signature pool helps the firewalls to instantly identify different bad actors. Almost all Web app firewalls (WAFs) can be customised for specific use cases and security policies in order to combat emerging threats. Finally, most of the modern solutions leverage the behaviour and reputational data to gain additional insights into the incoming traffic. WAFs are also integrated with other security solutions in order to form a security perimeter. These include Distributed Denial of Service (DDoS) protection to provide the additional scalability required to block a high-volume attack.

Best practices for securing a Web application

- Manually review the application to identify entry points and client-side codes. Classify the third-party hosted content separately to avoid any confusion.
- Test the application for all possible path traversals, horizontal and vertical access control issues, missing authorisation or any insecure, direct object references.
- Switch off all the unnecessary functionalities of the application.
- Always keep the remote access limited and secure, as hackers try to enter the application through this.
- Secure all the data transmissions involved in different parts of the Web application.
- Use accounts with limited privileges and permissions.
- Segregate the environments for development, testing and live usage of the Web application. Avoid testing on the development environment.
- Always install all the security patches and keep them updated. Always monitor and audit the server logs.
- Improve the application's resilience against Denial of Service threats by testing it perfectly for account lockout, anti-automation, HTTP protocol and SQL wildcard DoS.



Figure 4: Some of the common causes of security breaches in Web applications in 2018 (Image source: [googleimages.com](https://www.googleimages.com))

Points to ponder

As per the 2017 Trustwave Global Security Report:

1. The mean number of vulnerabilities associated with a Web application is 11.
2. The median number of days required to detect an application intrusion dropped to almost half in 2017 from that of 2016. Some of the internally-detected breaches were found in about 16 days itself.
3. Geographically, North America has been hit most with 49 per cent of all the data breaches occurring in 2017. The Asia Pacific region claimed 21 per cent of all breaches, while 20 per cent happened in Europe, the Middle East and Africa, and the remaining 10 per cent in Latin America.
4. The retail sector was the most hit industry, accounting for almost 22 per cent of the breaches. The point-of-sale breaches associated with the retail sector accounted for most of these breaches.
5. Breach incidents in e-commerce dropped by 12 per cent in 2017 as compared to 2016 due to security solutions implemented in the e-commerce sector.
6. In 2017, 35 per cent of the spam messages contained malware, which is actually a dramatic increase from the 3 per cent in 2016. And around 60 per cent of all inbound email tested was found to be spam. **END** 🐧

References

- [1] <http://www.wikipedia.org/>
- [2] <https://ieeexplore.ieee.org>
- [3] <https://www.networkworld.com/>
- [4] <https://www.itproportal.com>
- [5] <https://www.netsparker.com>

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